Project Plan

WBS, network diagram with critical path, Gantt chart with milestones and resource allocation

1. Summary

Schedule and effort come from the Velocity & Release Forecast: **96 story points** delivered across **8 two-week sprints (16 weeks)** at an average velocity of **12.67 story points/sprint** (observed velocities: 11, 14, 13 SP), followed by **3 weeks of System QA & 46-club pilot UAT** and **3 weeks of production rollout & junior maintainer handover** (**22 weeks total**). This leaves a **14-to-17-week safety buffer** before the volunteer student developers graduate at Month 9, within the **Rs. 2.4 lakh 3-year hosting budget (Rs. 0 developer cost)**.

2. Work Breakdown Structure

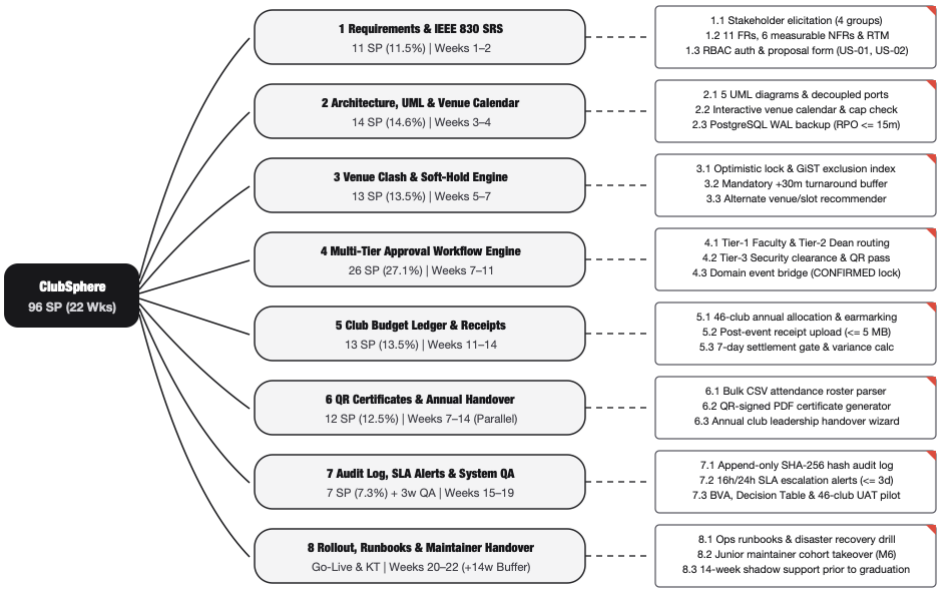


Figure 1 - WBS with story-point and schedule allocation (sums to the 96 story-point backlog across 8 sprints plus QA and handover).

WBS	Package	Story Points	Dur.	Pred.	ES / EF	LS / LF	Slack / Path
1.0	Requirements elicitation, IEEE 830 SRS & RBAC auth (Sp 1)	11 SP (11.5%)	2 wks	—	W0 / W2	W0 / W2	0w (CRITICAL)
2.0	UML architecture, DB schema & venue calendar (Sp 2)	14 SP (14.6%)	2 wks	1.0	W2 / W4	W2 / W4	0w (CRITICAL)
3.1	Venue soft-hold lock, +30m buffer & clash engine (Sp 3)	13 SP (13.5%)	3 wks	2.0	W4 / W7	W4 / W7	0w (CRITICAL)
3.2	3-tier approval workflow engine & domain events (Sp 4–5)	26 SP (27.1%)	4 wks	3.1	W7 / W11	W7 / W11	0w (CRITICAL)
3.3	Club budget ledger & 7-day receipt audit gate (Sp 6–7a)	13 SP (13.5%)	3 wks	3.2	W11 / W14	W11 / W14	0w (CRITICAL)
3.4	Annual club leadership handover wizard (FR-10)	6 SP (6.25%)	3 wks	2.0	W4 / W7	W11 / W14	+7w Float
3.5	Bulk CSV & QR participation certificate engine (FR-09)	6 SP (6.25%)	3 wks	3.1	W7 / W10	W11 / W14	+4w Float

WBS	Package	Story Points	Dur.	Pred.	ES / EF	LS / LF	Slack / Path
4.1	SHA-256 immutable audit log & 3-day SLA alerts (Sp 8)	7 SP (7.3%)	2 wks	3.3, 3.4, 3.5	W14 / W16	W14 / W16	0w (CRITICAL)
4.2	Operations runbooks & maintainer handover docs	Docs Pack	3 wks	3.3	W14 / W17	W16 / W19	+2w Float
5.0	System QA (BVA, Decision Tables) & 46-club UAT pilot	QA Suite	3 wks	4.1	W16 / W19	W16 / W19	0w (CRITICAL)
6.0	Production go-live & junior maintainer KT takeover	Release v1.0	3 wks	4.2, 5.0	W19 / W22	W19 / W22	0w (CRITICAL)
Total	Complete Scope + 14-Week Pre-Graduation Buffer (W22–W36)	96.0 SP (100%)	22w (+14w)	—	W0 / W22	W0 / W22	Critical = 22w

3. Network Diagram & Critical Path

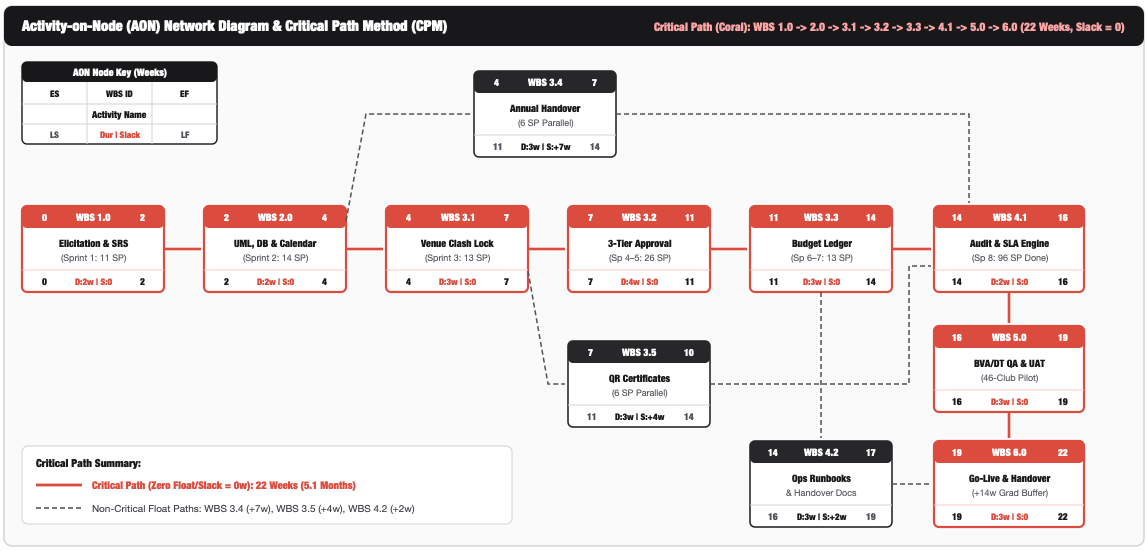


Figure 2 - Activity-on-Node (AON) network diagram. Coral nodes highlight the 22-week zero-slack Critical Path.

4. Gantt Chart & Milestone Gates

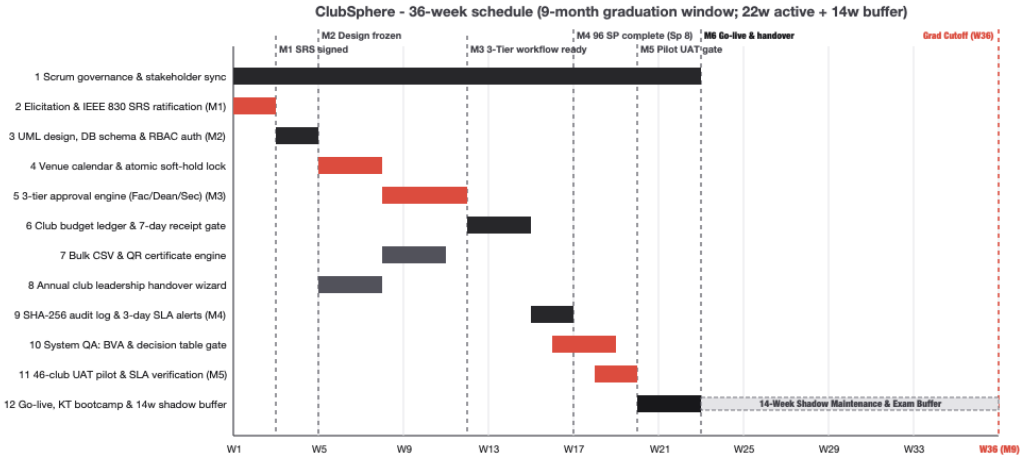


Figure 3 - Schedule across the 36-week (9-month) window. Coral bars mark stakeholder-critical and critical-path gates.

Milestone	Week	Gate & Verification Criteria
M1 IEEE 830 SRS ratified	2	Club leaders, Faculty, Dean's Office and Security Desk sign off on 11 FRs & 6 measurable NFRs; Sprint 1 (11 SP) live
M2 Architecture frozen	4	5 UML diagrams baselined; decoupled IVenueBookingPort contract frozen; automated PostgreSQL WAL backup active
M3 3-Tier workflow & clash gate	11	Zero-clash GiST lock (+30m buffer) and Faculty -> Dean -> Security domain event pipeline pass integration suite
M4 96 SP feature complete	16	All 96 story points completed at Sprint 8 (mean velocity 12.67 SP/sprint); SHA-256 audit log & SLA alerts active
M5 46-club pilot UAT gate	19	All BVA & Decision Table tests green (DRE = 95%); 46 club leaders complete pilot dry-run (mean proposal time 2m 48s)
M6 Go-live & handover complete	22	Junior Maintainers (2nd/3rd-yr devs) pass solo deployment & DB recovery drill 14 weeks before seniors graduate

5. Scrum Team Resource Allocation & 3-Year Budget (Rs. 2.4 Lakh)

Role / Cost Item	Assigned Personnel / Specification	Commitment / Cost	Responsibility & Handover Role
Product Owner (PO)	Associate Dean of Student Affairs	6 hrs/wk (Rs. 0)	Owns 96-SP backlog, MoSCoW scope gate, and ≤ 3-day SLA policy
Scrum Master & QA Lead	Senior Student Dev-1 (Final Year)	15 hrs/wk (Rs. 0)	Facilitates sprints, leads BVA/Decision Table QA, authors ops runbooks
Tech Lead & Backend Dev	Senior Student Dev-2 (Final Year)	18 hrs/wk (Rs. 0)	Builds ApprovalWorkflowEngine, GiST venue lock, and SHA-256 audit trail
Full-Stack & Frontend Devs	Senior Student Dev-3 & Dev-4 (Final Year)	17 hrs/wk each (Rs. 0)	Builds BudgetLedger, QR certificates, Proposal Wizard, and Handover UI
Shadow Maintainers (Handover)	Junior Student Devs Jr-1 & Jr-2 (2nd/3rd Yr)	8 hrs/wk each (Rs. 0)	Co-review PRs from Sprint 3; take over primary maintenance at Week 22 (M6)
Infra & Budget Custodian	Campus IT System Administrator	Rs. 2.40 Lakh / 3 Yrs	Cloud VPS (Rs. 1.01L), S3 WAL Backups (Rs. 39.6k), SMTP/SSL/CDN (Rs. 75.6k), Reserve (Rs. 24k)